From reference to manufacturing handover

Working example: 993 Turbo carrier · current master: 1.274798 kg · limit: 1.30 kg

Images + reference scale

3D scan + units + orientation

Drawing + axes + dimensions

1 Editable geometry

build123d / OCCT · PicoGK

KEEP

Source, parameters and CAD master

Check axes, dimensions, solid, mass
Record adopted dimensions

2 Surface and triangles

PyMesh · Gmsh · CGAL · libigl

KEEP

Revised STL and before/after report

Closure, intersections and quality
Measure distance to the CAD master

3 Mechanical calculations

Gmsh C3D10 · CalculiX

KEEP

Loads, mounts, material and fields

Compare stress and displacement
Audit energy and mesh sensitivity

4 Orientation and supports

PySLM · PyMesh / CGAL

KEEP

Pose, envelopes and actual surfaces

Coverage, contacts and confinement
Retain evidence and verdicts

5 Layers and scan vectors

PySLM · sections and exact audits

KEEP

Every layer and every scan vector

Audit whole segments and holes
Verify actual exposed contacts

6 Manufacturing handover

Offline guide · PDF · manifest

DELIVER

Drawings, CAD/STL, supports, layers

Results, settings and reproduction
Assign machine format and laser law

After printing: workshop report → corrections → a controlled new revision

Workflow map · stage results remain in their own reports · current geometry snapshot: 18 October 2026